

Andrew “AJ” Jones

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EDUCATION

University of Notre Dame | Notre Dame, IN

Graduate: May 2027

Bachelor of Science in Engineering

Major: Computer Science | Minor: Engineering Corporate Practice

EXPERIENCE

Make a Play D - League Fellow '25 | Remote

January 2025 - May 2025

- Competed individually against 200+ high-achieving college student-athletes in weekly mock interview simulations, honing professional communication, presentation, and interviewing skills
- Partnered with Oliver Wyman consultants and a case team to develop a comprehensive 30-year growth strategy for future airport operations
- Designed elements of a strategic plan aiming to transform airports by 2055, focusing on net-zero transitions, technological innovation, and enhanced passenger experience.
- Contributed to designing a future vision for airports as personalized, seamless, and operationally efficient ecosystems by 2055.

Undergraduate Research Assistant - Machine Intelligence Lab | Notre Dame, IN

September 2024 – Present

- Developed an automated tool using Python and Large Language Models (LLMs) to extract thermal performance curves (TPCs) from scientific literature.
- Enhanced data collection efficiency by reducing manual extraction time and ensuring scalability with ongoing publications.
- Collaborated with faculty to streamline data pipelines and validate extracted information against existing datasets.

Undergraduate Research Assistant - Security and Software Engineering Lab | Notre Dame, IN

May 2024 – July 2024

- Participated in a research project aimed at benchmarking the capabilities of code generation models (e.g., GitHub Copilot, ChatGPT, Gemini) in developing secure and functional software. The project involved generating datasets to assess OWASP compliance and the effectiveness of language models in merging security-related pull requests.
- Developed a comprehensive understanding of secure coding practices and source code vulnerabilities.
- Enhanced capabilities of code generation models through rigorous testing and benchmarking.

Fleet Feet | Omaha, Nebraska

August 2022 – Present

- Educated customers about running shoes with excellent customer service
- Utilized product knowledge to provide in-depth information about various shoe brands, technologies, and accessories
- Performed over 50 gait analyses per week, improving customer shoe recommendations by 3%

LEADERSHIP AND ACTIVITIES

Student-Athlete | Notre Dame, IN

Track & Field

August 2023 – Present

- Competed as a member of the varsity track and field team, balancing a rigorous training schedule with academic responsibilities.
- Demonstrated strong time management and organizational skills by maintaining academic excellence while committing 20+ hours per week to practice, travel, and competition.

RELATED COURSEWORK OR PROJECTS

CSE 20312 Data Structures | University of Notre Dame

January 2025 – May 2024

- Developed software utilizing dynamic arrays, linked lists, stacks, queues, hash tables, and binary trees
- Analyzed computational complexity and implemented search and sorting algorithms
- Applied object-oriented programming principles and recursion to design efficient data-driven applications

CSE 20289 Systems Programming | University of Notre Dame

January 2025 – May 2025

- Navigated Unix-based systems, managed files and processes, and explored system and network resources with command-line utilities
- Developed shell scripts combining Unix commands to automate tasks and handle complex pipelines.
- Wrote regular expressions to filter and process data, and utilized debugging and profiling tools to enhance software efficiency.
- Built software using system calls in C, managing memory, file systems, and socket communication.

TECHNICAL AND LANGUAGE SKILLS

Programming: Python, C, MatLab, Excel, Solid Works

Data Handling: Pandas, CSV, JSON, SQL

APIs: GitHub REST API, Stack Overflow API, Python “requests” module

Software Security: OWASP compliance, vulnerability detection

Machine Learning: Language model prompt engineering (OpenAI, Gemini)

Project Management: GitHub, Directory traversal, branch management, pull request, Microsoft teams

